we have seen:

$$\varphi(k) = [-y(k-1), \cdots, -y(k-n), u(k), \cdots, u(k-m)]^T$$

$$\Downarrow$$

$$(\circ) \quad \hat{y}(k|\hat{\theta}) = \varphi^T(k)\,\underline{\hat{\theta}} \qquad \underline{\hat{\theta}} = [\hat{a}_1 \cdots \hat{a}_n \; \hat{b}_0 \cdots \hat{b}_m]$$

what if we choose

$$\tilde{\varphi}(k) = [\,y(k-1), \cdots, y(k-n), u(k), \cdots, u(k-m)]^T$$

$$(\circ\circ) \quad \hat{y}(k|\hat{\theta}) = \tilde{\varphi}^T(k)\,\underline{\hat{\tilde{\theta}}} \qquad \underline{\hat{\tilde{\theta}}} = [-\hat{a}_1 \cdots -\hat{a}_n \; \hat{b}_0 \cdots \hat{b}_m]$$

$\Rightarrow$ $(\circ)$ & $(\circ\circ)$ are equiv.

$\Rightarrow$ We could choose the regressor in the second form.

$$\Rightarrow \quad \underline{\hat{\tilde{\theta}}}^* = (\tilde{\Phi}_N \tilde{\Phi}_N^T)\,\tilde{\Phi}_N\, y_N$$

where $\tilde{\Phi}_N = [\tilde{\varphi}(1) \cdots \tilde{\varphi}(N)]$

$$\Rightarrow \quad \underline{\hat{\tilde{\theta}}}^* = [-\hat{a}_1^*, \cdots, -\hat{a}_n^*, \hat{b}_0^*, \cdots, \hat{b}_m^*]^T$$

we have seen one shot sol. thanks to linearity in ~~the~~ $\underline{\theta}$

What to do if the model is NOT liner in $\underline{\theta}$

E.g. OE: $\hat{y}(k|\hat{\underline{\theta}}) = \dfrac{\hat{B}(z|\hat{\underline{\theta}})}{\hat{A}(z|\hat{\underline{\theta}})} u(k)$ $\qquad \hat{\underline{\theta}} = [\hat{a}_1, \dots, \hat{a}_n, \hat{b}_0, \dots, \hat{b}_m]^T$

we do not have the possibility of writing this as a linear combination of a finite number of parameters.

in fact $\hat{y}(k|\hat{\theta}) = [\hat{g}(0) + \hat{g}(1) z^{-1} + \hat{g}(2) z^{-2} \dots\dots]\, u(k)$ $\leadsto \infty$

long division.

$$= \hat{g}(0) u(k) + \hat{g}(1) u(k-1) + \hat{g}(2) u(k-2) + \dots\dots \leadsto \infty$$

$$= \underbrace{[u(k), u(k-1), u(k-2), \dots\dots]}_{\text{similar to the regressor}}{}^{T} \begin{bmatrix} \hat{g}(0) \\ \hat{g}(1) \\ \hat{g}(2) \\ \vdots \end{bmatrix}$$

similar to a param. vector

but infinitely long !!!

This would correspond to identification of all the terms of the impulse response of the system.

$\Rightarrow$ ∞ terms unless F.I.R. case

ARMAX $\quad \hat{y}(k|\underline{\hat{\theta}}) = \left[1 - \dfrac{\hat{A}(z|\hat{\theta})}{\hat{D}(z|\hat{\theta})}\right] y(k) + \dfrac{\hat{B}(z|\underline{\hat{\theta}})}{\hat{D}(z|\hat{\theta})} u(k)$

also here NOT linearity in $\underline{\hat{\theta}} = [\hat{a}_1 \dots \hat{a}_n, \hat{b}_0 \dots \hat{b}_m, \hat{d}_1 \dots \hat{d}_q]$ nor in any finite dimension set of other equivalent parameters.

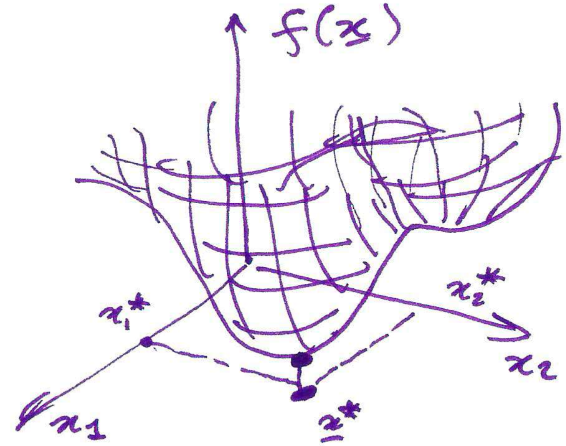
Recall: ARX: $\quad \hat{y}(k|\hat{\theta}) = [1 - \hat{A}(z|\underline{\hat{\theta}})]\, y(k) + \hat{B}(z|\underline{\hat{\theta}})\, u(k)$

ARX is linear in the parameters. No parameter in the denominators.

To identify not-"lin. in θ" models we will resort to iterative methods for minimization of $J_N(\underline{\hat{\theta}})$.

MATHEMATICAL PROGRAMMING

Algorithms for minimizing a scalar function of a vector argument, by starting from a temptative solution and changing the solution "step by step" on the basis of local informations.



Pb. $\underline{x}^* = \arg\min_{\underline{x}} f(\underline{x})$

An iterative method will start from

$\underline{x}^{(0)}$ = "temptative solution" (must be "reasonable")

for $l = 0, 1, 2, \ldots$

$\underline{x}^{(l+1)}$ = function of (local informations at $\underline{x}^{(l)}$)

e.g. : $f(\underline{x}^{(l)})$, $\left.\frac{\partial f(\underline{x})}{\partial \underline{x}}\right|_{\underline{x}=\underline{x}^{(l)}}$ $\ldots$, $\ldots$

with some stopping criteria.

Stopping criteria:

- $\|\underline{x}^{(l+1)} - \underline{x}^{(l)}\| \le \varepsilon_x$ for some "small" ε_x

- $|f(\underline{x}^{(l+1)}) - f(\underline{x}^{(l)})| \le \varepsilon_f$ " " " ε_f

- maximum number of iterations:

  $l \le L_{max}$ for a pre-determined value L_{max}

  ($\equiv$ lack of time)

  $\hookrightarrow$ until some T_{max} (time)

Different stopping criteria can be combined.

<u>Examples</u> of math. prog. methods.

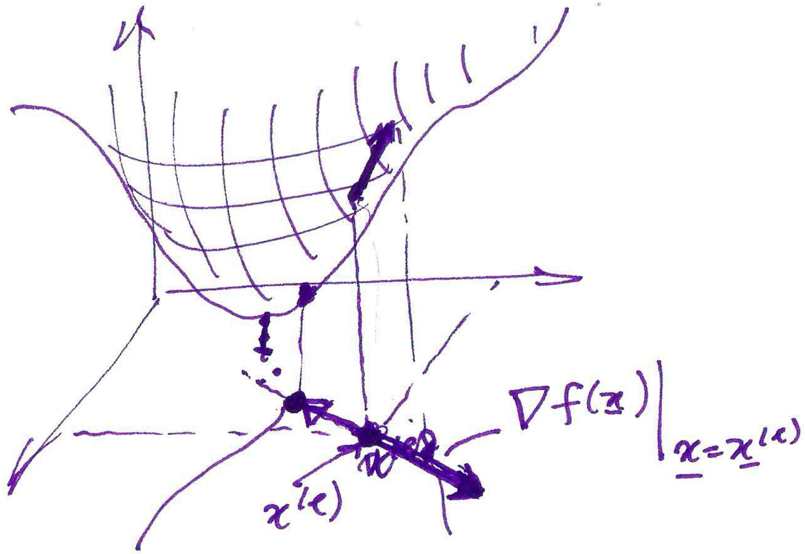
gradient descent: $\underline{x}^{(l+1)} = \underline{x}^{(l)} - \alpha^{(l)} \nabla f(\underline{x})\Big|_{\underline{x}=\underline{x}^{(l)}}$

step.

gradient $\nabla f(\underline{x}) = \underbrace{\left[\frac{\partial f(\underline{x})}{\partial x_1}, \ldots\ldots, \frac{\partial f(\underline{x})}{\partial x_p} \right]}_{\frac{\partial f(\underline{x})}{\partial \underline{x}} \in \mathbb{R}^{1\times p}}^T \in \mathbb{R}^{p\times 1}$

(for dim $\underline{x} = p$)

$$\nabla f(\underline{x}) \equiv \left[\frac{\partial f(\underline{x})}{\partial \underline{x}} \right]^T$$



$$x^{(l+1)} = x^{(l)} - \alpha^{(l)} \nabla f(\underline{x})\big|_{\underline{x}=\underline{x}^{(l)}}$$

$\alpha^{(l)} = \text{constant} = \alpha$

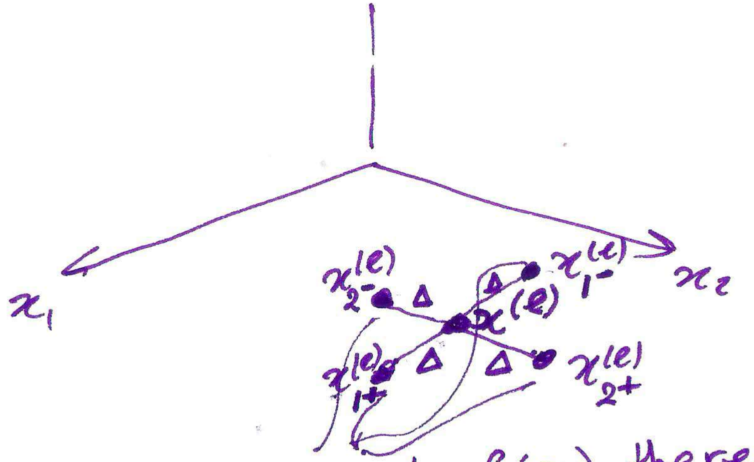
other possibilities:

$\alpha^{(l)} = \frac{c_1}{c_2 + l}$ decreases with l

gradient method is of "order 1" (or type)

(makes use of $\frac{\partial f(\underline{x})}{\partial \underline{x}}$)

~~Assum~~ If $\frac{\partial f(\underline{x})}{\partial \underline{x}}$ is not available, but only $f(\underline{x})$ can be computed $\Rightarrow$ methods of "order 0"



compute $f(\underline{x})$ there $\Rightarrow$ use to approxim. the gradient.

$$x_{1-}^{(\ell)} = \underline{x}^{(\ell)} - \begin{bmatrix} \Delta \\ 0 \end{bmatrix}$$

$$x_{1+}^{(\ell)} = \underline{x}^{(\ell)} + \begin{bmatrix} \Delta \\ 0 \end{bmatrix}$$

$\vdots$

$$\nabla f(\underline{x})\Big|_{\underline{x}=\underline{x}^{(\ell)}} \cong \begin{bmatrix} \dfrac{f(x_{1+}^{(\ell)}) - f(x_{1-}^{(\ell)})}{2\Delta} \\ \\ \dfrac{f(x_{2+}^{(\ell)}) - f(x_{2-}^{(\ell)})}{2\Delta} \end{bmatrix}$$

$\Rightarrow$ apply an approximate gradient method

If we can use $f(\underline{x})$, $\frac{\partial f(\underline{x})}{\partial \underline{x}}$, & $\frac{\partial^2 f(\underline{x})}{\partial \underline{x}^2}$ $\Rightarrow$ meth. of "order 2" (I 90

Consider $f(\underline{x})$ and its second order Taylor approx. around $\underline{x}^{(\ell)}$

$$f(\underline{x}) \cong f(\underline{x}^{(\ell)}) + \left.\frac{\partial f(\underline{x})}{\partial \underline{x}}\right|_{\underline{x}=\underline{x}^{(\ell)}} (\underline{x}-\underline{x}^{(\ell)}) + \frac{1}{2}(\underline{x}-\underline{x}^{(\ell)})^T \left.\frac{\partial^2 f(\underline{x})}{\partial \underline{x}^2}\right|_{\underline{x}=\underline{x}^{(\ell)}} (\underline{x}-\underline{x}^{(\ell)}) \cancel{+ \text{Rest}}$$

$$= \text{function of } (x - x^{(\ell)})$$

Consider.

$$\text{func}(x-x^{(\ell)}) = \frac{1}{2}(x-x^{(\ell)})^T M_\ell (x-x^{(\ell)}) - c^T(x-x^{(\ell)})$$

$$\text{where } M_\ell = \left.\frac{\partial^2 f(\underline{x})}{\partial \underline{x}^2}\right|_{\underline{x}=\underline{x}^{(\ell)}} \quad \text{and} \quad -c^T = + \left.\frac{\partial f(\underline{x})}{\partial \underline{x}}\right|_{\underline{x}=\underline{x}^{(\ell)}}$$

it is a quadratic form.

in $\underline{z} = (x - x^{(\ell)})$ $\quad \text{func}(\underline{z}) = \frac{1}{2}\underline{z}^T M_\ell \underline{z} - c^T \underline{z}$

if $M_\ell > 0 \Rightarrow \text{func}(\underline{z})$ has a minimum in.

(191)

$$\underline{z}^* = M_\ell^{-1}\underline{c} \Rightarrow (\underline{x}-\underline{x}^{(\ell)})^* = -\left[\frac{\partial^2 f(\underline{x})}{\partial \underline{x}^2}\bigg|_{\underline{x}=\underline{x}^{(\ell)}}\right]^{-1}\left[\frac{\partial f(\underline{x})}{\partial \underline{x}}\bigg|_{\underline{x}=\underline{x}^{(\ell)}}\right]^T$$

$$(\underline{x}-\underline{x}^{(\ell)})^* = \underline{x}_\ell^* - \underline{x}^{(\ell)}$$

$$\text{if } \underbrace{\frac{\partial^2 f(\underline{x})}{\partial x^2}\bigg|_{\underline{x}=\underline{x}^{(\ell)}}}_{\triangleq H(x^{(\ell)}) \text{ "the Hessian.}} > 0 \Rightarrow \underline{x}_\ell^* = \underline{x}^{(\ell)} - \left[\frac{\partial^2 f(\underline{x})}{\partial \underline{x}^2}\bigg|_{\underline{x}-\underline{x}^{(\ell)}}\right]^{-1}\underbrace{\left[\frac{\partial f(\underline{x})}{\partial \underline{x}}\bigg|_{\underline{x}=\underline{x}^{(\ell)}}\right]^T}_{\equiv \nabla f(\underline{x})|_{\underline{x}=\underline{x}^{(\ell)}}}$$

$$\equiv \quad \text{if } H(x^{(\ell)}) > 0 \Rightarrow x^{(\ell+1)} = x^{(\ell)} - [H(x^{(\ell)})]^{-1} \nabla f(\underline{x})\Big|_{\underline{x}=\underline{x}^{(\ell)}}$$

Note that if $f(\underline{x})$ is a quadratic form with. $M = H(\underline{x}^{(\ell)}) > 0 \ \forall \underline{x}^{(\ell)}$

$\Rightarrow$ this alg. converges in one step.

in general $H(x^{(\ell)})$ is not always pos. definite

$\Rightarrow$ we modify the alg.

$$\underline{x}^{(\ell+1)} = \underline{x}^{(\ell)} - \alpha^{(\ell)} \left[H(\underline{x}^{(\ell)}) + M^{(\ell)} \right]^{-1} \nabla f(\underline{x}) \Big|_{\underline{x}=\underline{x}^{(\ell)}}$$

$$M^{(\ell)} \text{ s.t. } \left[H(x^{(\ell)}) + M^{(\ell)} \right] > 0$$

Gauss-Newthon method.

$$H(\underline{x}^{(\ell)}) > 0 \Rightarrow \begin{cases} M^{(\ell)} = 0 \quad \text{i.e.:} \\ \underline{x}^{(\ell+1)} = x^{(\ell)} - \alpha^{(\ell)} \left[H(\underline{x}^{(\ell)}) \right]^{-1} \nabla f(\underline{x}) \Big|_{\underline{x}=\underline{x}^{(\ell)}} \end{cases}$$

$$\text{if } H(x^{(\ell)}) \not> 0 \Rightarrow \begin{cases} M^{(\ell)} = I - H(x^{(\ell)}) \\ x^{(\ell+1)} = x^{(\ell)} - \alpha^{(\ell)} \nabla f(x) \Big|_{\underline{x}=\underline{x}^{(\ell)}} \end{cases}$$

That is: if $H(x^{(\ell)})$ is not ">0" $\Rightarrow$ only one step of the gradient method.